

Report on the use of passive acoustic monitoring for songbirds and seabirds on Lanz and Cox Islands

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 Note

This report is dynamically generated, meaning its results may evolve with the addition of new data or further analyses. For the most recent updates, refer to the publication date and feel free to reach out to the authors.

1 Abstract

Passive acoustic monitoring was conducted on Lanz and Cox Islands to characterize seabird and songbird communities. Autonomous Recording Units (ARUs; SM4) were deployed at 0 locations

and analyzed using a combination of machine-learning algorithms and expert validation. This report summarizes species presence, temporal patterns of detection, and relative abundance for seabirds and songbirds, and provides spatial representations of detections across both islands. These results establish a baseline to support evaluation of seabird prospecting activity and future changes in avian communities.

2 Land Acknowledgement

3 Introduction

Lanz and Cox Islands support important seabird and songbird habitats, yet populations have been negatively affected by the presence of invasive predators, including mink and raccoon. As part of a broader restoration initiative, passive acoustic monitoring was implemented to document avian community composition prior to eradication activities. Autonomous recording units (ARUs) were deployed across both islands over three breeding seasons to capture vocal activity of seabirds and songbirds. Acoustic data were processed using the WildTrax data management platform in combination with BirdNET and HawkEars machine-learning algorithms, with detections verified using expert review informed by prior experience in marine acoustic environments. This report presents the results of these analyses, including species detected, timing of detections, relative abundance metrics, and spatial patterns of occurrence. Together, these data provide a baseline against which post-eradication monitoring can be compared to assess project outcomes and ecological recovery.

4 Methods

4.1 Data collection

ARUs were deployed in three stages between 2024 and 2025 across Lanz and Cox Islands (Figure 1), with the final data retrievals taking place in the winter of 2025-26. A total of 129243 recordings were collected totalling 10736 audio hours of data (Figure 2).

Table 1: Locations surveyed across years. Ones indicated a deployment in that year for that location

location	latitude	longitude	2024	2025
DC01	50.80230	-128.6360	1	1
DC02	50.80143	-128.6338	1	1
DC03	50.78923	-128.6254	1	0
DC06	50.80157	-128.5735	0	1
DC07	50.80359	-128.6167	1	1
DC09	50.79729	-128.6203	1	1
DC10	50.79253	-128.6171	1	1
DC11	50.81005	-128.6068	1	1
DC13	50.80575	-128.6379	1	1
DC15	50.79417	-128.6310	1	1

Table 1: Locations surveyed across years. Ones indicated a deployment in that year for that location

location	latitude	longitude	2024	2025
DL01	50.82893	-128.7070	1	1
DL02	50.81575	-128.7114	1	1
DL05	50.80633	-128.6610	1	1
DL06	50.82519	-128.7040	1	1
DL07	50.81814	-128.7049	1	1
DL09	50.81645	-128.7001	1	1
DL10	50.82627	-128.6822	1	1
DL11	50.81652	-128.6691	0	1
DL12	50.81344	-128.6660	1	1
DL15	50.82689	-128.7106	1	1

4.2 Data management, processing and quality control

All recordings were uploaded to [WildTrax](#) in order for the data to be accessible and stored. Locations name were confirmed against the deployment information to ensure data from each location was consistently named across years.

4.3 Community data processing

For each sampling location and year, a total of nine autonomous recording unit (ARU) recordings, each 3 minutes in duration, were analyzed to characterize the songbird community. All vocalizing species were identified, and the abundance of each species was estimated using a count-removal framework based on time-to-first-detection, which helps reduce positive bias associated with repeated detections of the same individual within a recording. Recordings were intentionally distributed across diel periods to capture variation in vocal activity. Five recordings were selected during the dawn period (04:00–07:59), when songbird detectability is typically highest. Two recordings were selected during dusk (19:00–22:59), and two during the night (23:00–03:59), allowing for detection of crepuscular or nocturnally active species and providing a more complete representation of the avian community. This standardized temporal sampling design ensured consistent effort across locations and years while maximizing detectability across species with differing activity patterns. Coastal proximity may reduce ARU detection effectiveness, as wave action and wind-driven noise can mask vocalizations and shrink effective detection distances (Pijanowski et al. (2011)). We therefore examined whether distance to the nearest coastline predicted observed species richness across monitoring locations. We also compared geophonic acoustic indices (Towsey et al. (2017)) against WildTrax noise metrics to establish thresholds for task exclusion and to characterize background noise conditions at each site.

4.4 Evaluating the performance of HawkEars v1.0.8 on seabirds

To assess the accuracy of the HawkEars classifier for three focal species, Common Murre, Marbled Murrelet, and Black Oystercatcher, we extracted all recordings from WildTrax in which the

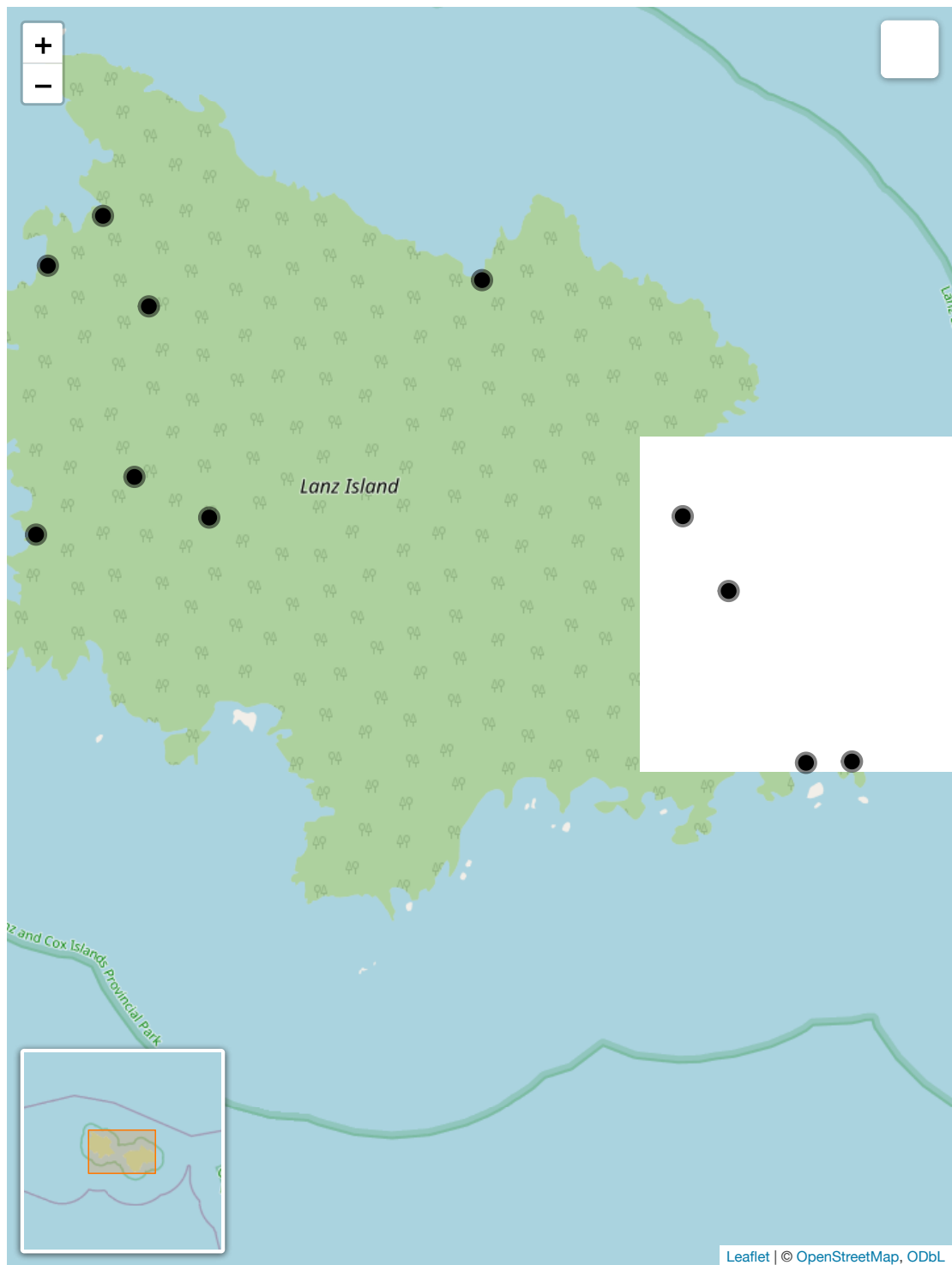


Figure 1: Locations from Lanz and Cox Provincial Park ARU Monitoring Program

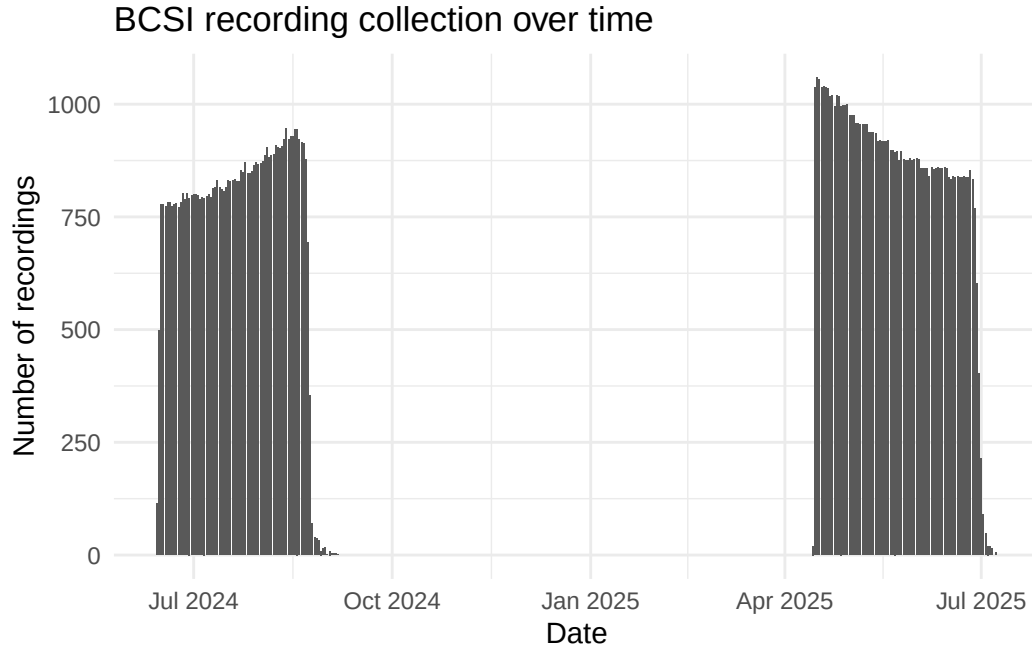


Figure 2: Recordings collected from Lanz and Cox Provincial Park ARU Monitoring Program

classifier returned a detection score 0.5. A human reviewer then verified each flagged task, confirming detections as true positives (TP) where the species was audible, or false positives (FP) where no supporting vocalization could be identified. Using these verified results, we applied the `wt_evaluate_classifier()` and `wt_classifier_threshold()` functions from the `wildtrax` R package to calculate precision, recall, and F-score across a range of score thresholds. This allowed us to identify the minimum confidence threshold at which classifier performance was deemed acceptable for each species — balancing the trade-off between retaining true detections and excluding false positives. Recordings scoring below this optimized threshold were excluded from subsequent analyses, while those meeting or exceeding it were accepted without further manual review. This approach is consistent with emerging best practices for deploying automated classifiers at scale, where full human review is impractical and a validated threshold provides a reproducible and defensible basis for data inclusion.

```
# A tibble: 2 x 2
```

	classifier	threshold
	<chr>	<dbl>
1	Birdnet v2.1	0.53
2	Hawkears v1.0.8	0.4

```
# A tibble: 11 x 14
```

	location_id	recording_id	task_id	species_common_name	confidence	organization
	<int>	<dbl>	<dbl>	<chr>	<dbl>	<chr>
1	557160	3599850	3364866	Marbled Murrelet	0.802	BCSI
2	557160	3601274	3364879	Black Oystercatcher	0.819	BCSI
3	557160	3644085	3369077	Common Murre	0.806	BCSI

4	557160	3644739	3369083	Marbled Murrelet	0.803	BCSI
5	557160	3646079	3369094	Marbled Murrelet	0.878	BCSI
6	557957	3609580	3364898	Common Murre	0.939	BCSI
7	557959	3617005	3364920	Marbled Murrelet	0.816	BCSI
8	557959	3617842	3364926	Marbled Murrelet	0.884	BCSI
9	557976	3380203	3364840	Black Oystercatcher	0.988	BCSI
10	557976	3528851	3364841	Black Oystercatcher	0.935	BCSI
11	557979	3612564	3364913	Marbled Murrelet	0.813	BCSI

```
# i 8 more variables: version <chr>, location <chr>,
#   recording_date_time <dtm>, recording_length <dbl>, scientific_name <chr>,
#   species_code <chr>, ai_detection_time <dbl>,
#   is_species_allowed_in_project <lgl>
```

4.5 HawkEars 2.0

Insert HawkEars 2.0 here.

5 Results

5.1 Species richness and diversity

A total of 36 species were found. Figure 3 describes the relationship of species richness for each location across the two years of surveys. Shannon's diversity was also calculated (see Figure 4).

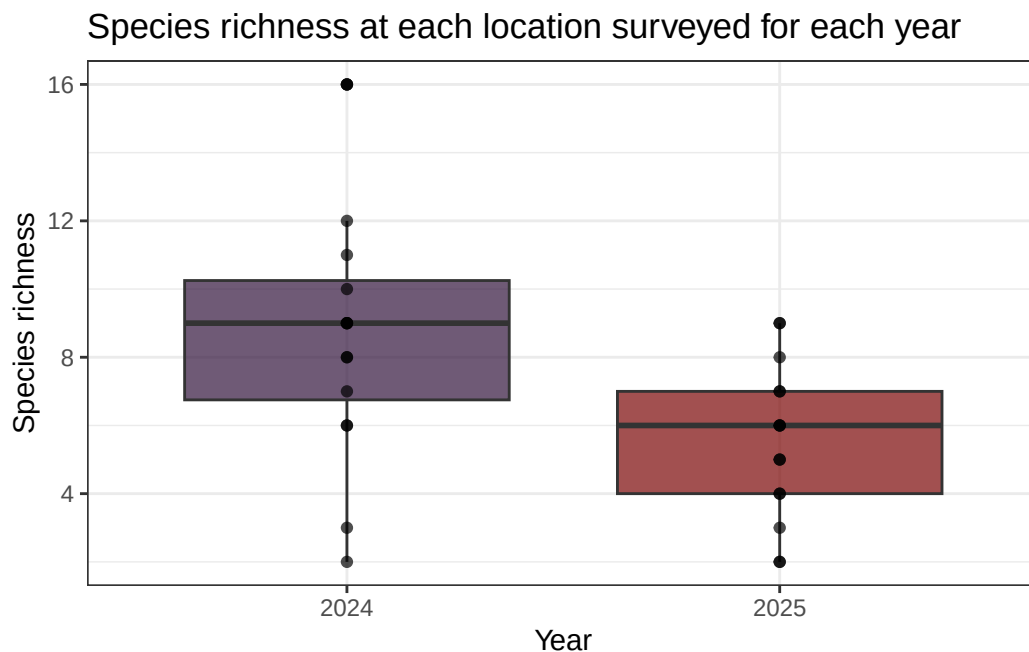


Figure 3: Species richness at forest monitoring locations across years

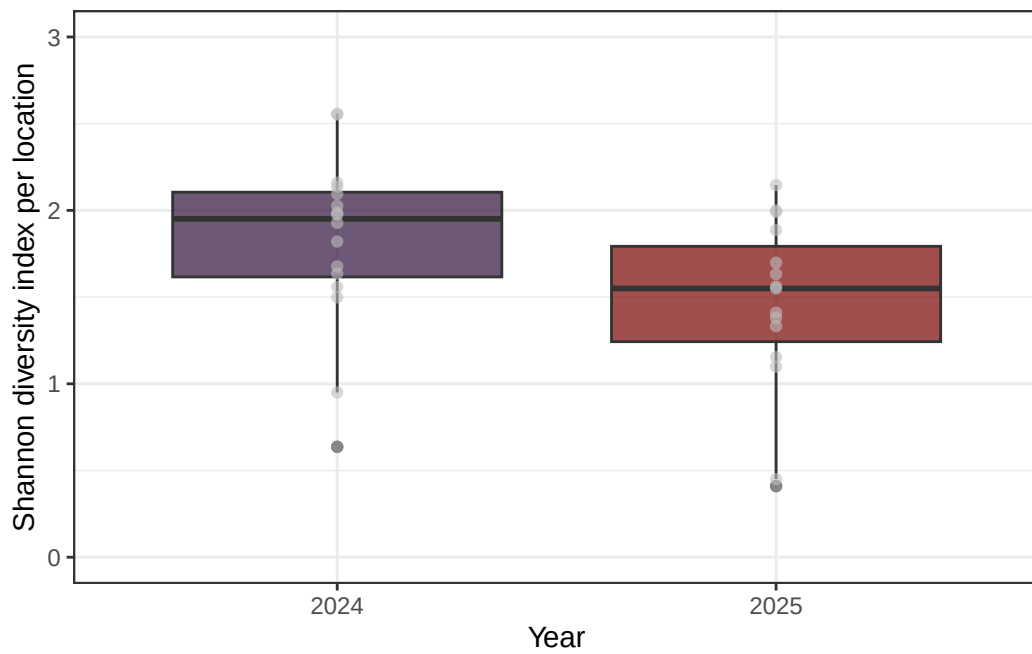


Figure 4: Shannon diversity index over years

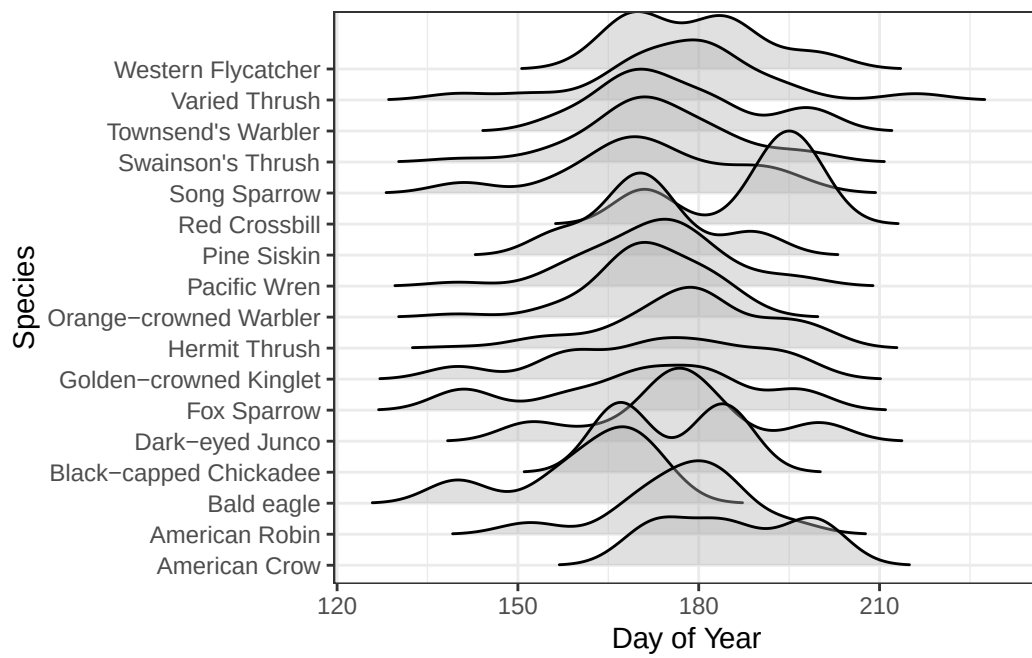
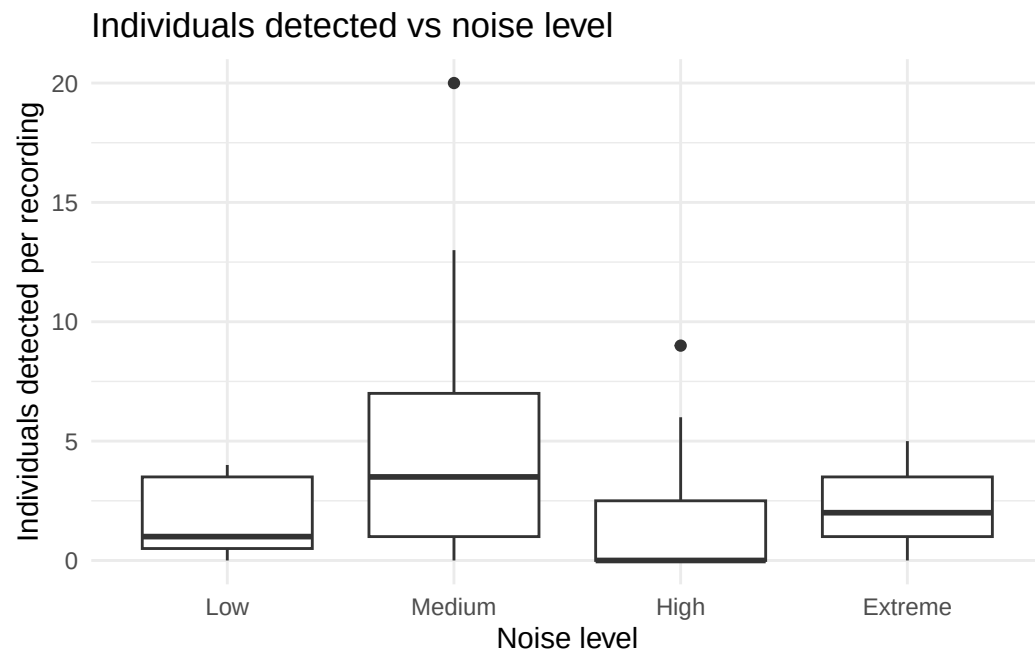
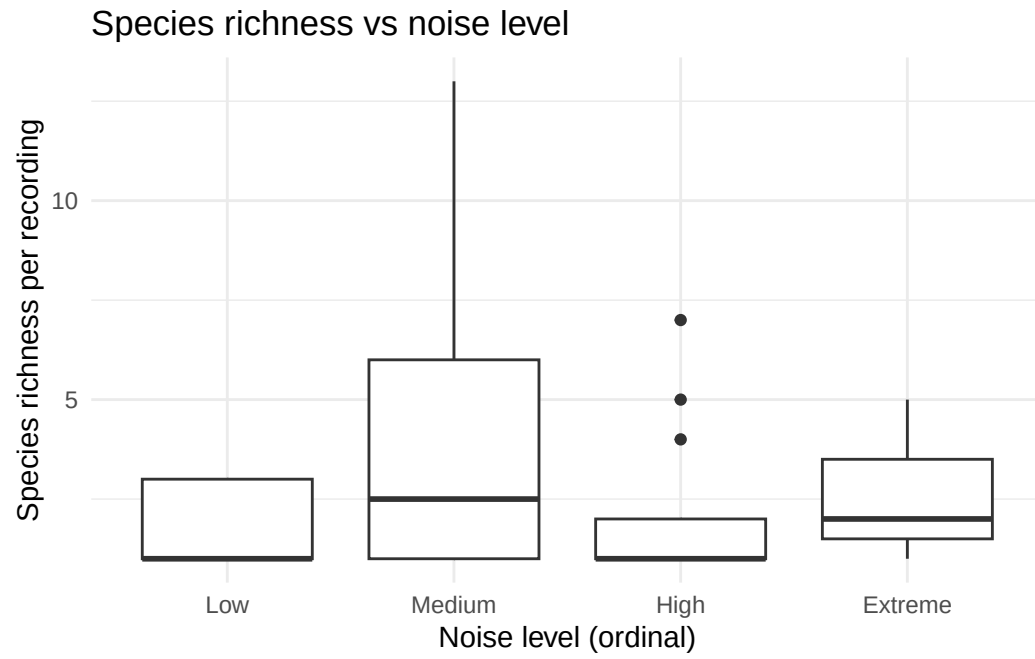
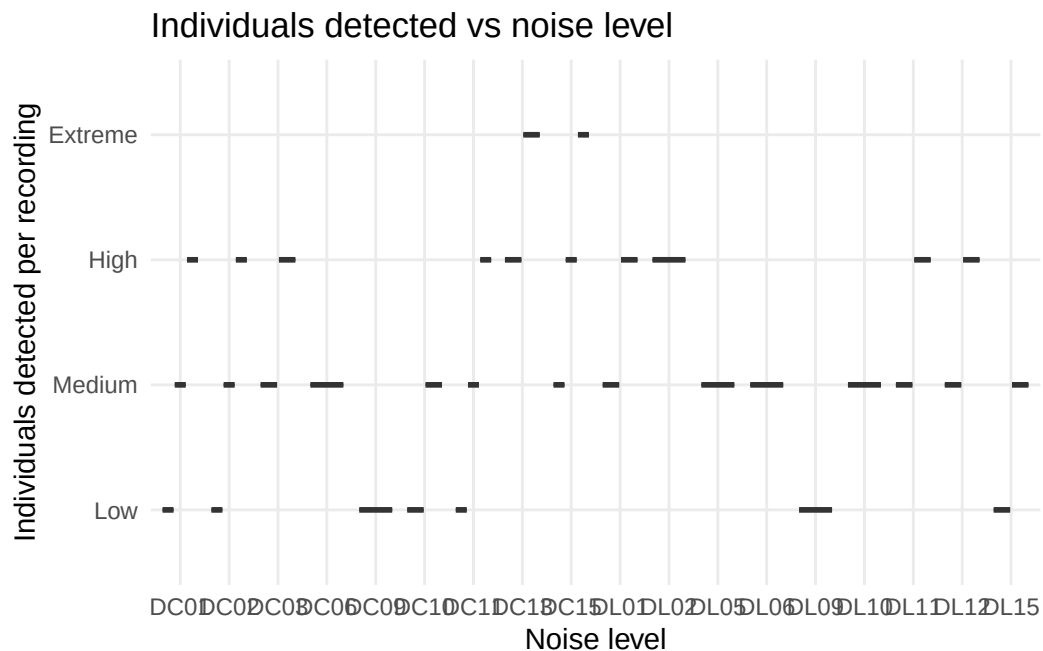


Figure 5: Seasonal detection activity of most commonly detected forest species

5.1.1 Proximity to ocean and noise effects





Contrary to expectations, distance to the nearest coastline showed no meaningful association with species richness across ARU monitoring locations (see [?@fig-ocean-richness](#)). Species richness was highly variable at all distances, ranging from approximately 3–32 species regardless of coastal proximity, and the fitted regression line was nearly flat across the full range of distances sampled (0–2, 200 m). This suggests that, within PEINP, coastal noise does not appear to systematically suppress acoustic detectability or species richness at the distances represented by our monitoring network, though the majority of stations were clustered within 500 m of the coastline, limiting inference at greater distances.

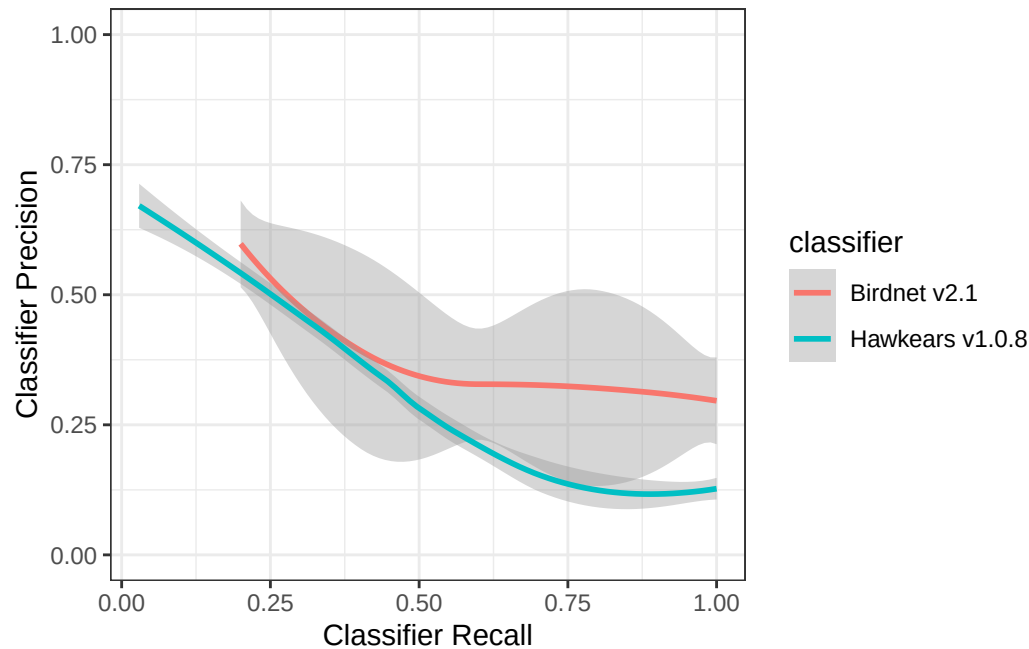
5.2 Classifier results

5.2.1 Evaluation of HawkEars v1.0.8

We showed that ...

```
ggplot(eval, aes(x=recall, y=precision, colour=classifier)) +
  geom_smooth() +
  theme_bw() +
  xlab("Classifier Recall") +
  ylab("Classifier Precision") +
  ylim(0,1)
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



5.2.2 HawkEars 2.0

Discussion and recommendations

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